

SRMSS — Smart Route Management & Scheduling System

Project Timeline & Milestones | Web application for public-transport depot operations

18 May 2026 → 16 June 2026 • ~4.5 weeks

METHODOLOGY

Iterative SDLC — Research → Plan → Design → Build → Test → Fix → Deploy

TECH STACK

PHP 8.4 · Laravel 13 · Livewire 4 · Flux UI · Tailwind 4 · DomPDF

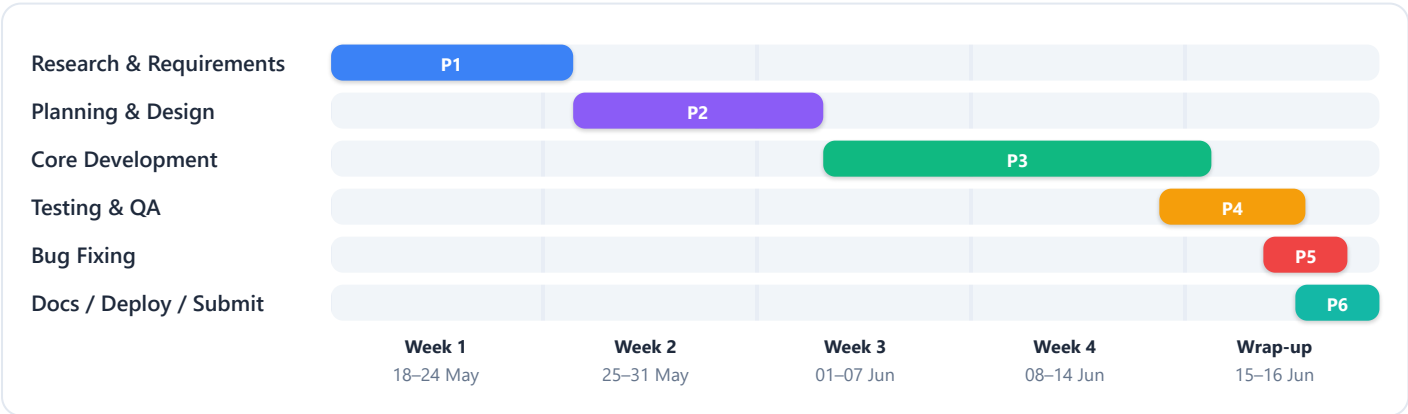
ROLES

Admin · Supervisor · Operator · Driver

1 Phase Overview

Phase	Activity	Dates	Duration
P1	Research & Requirements	18 May – 24 May	1 week
P2	Planning & System Design	25 May – 31 May	1 week
P3	Core Development	01 Jun – 11 Jun	1.5 weeks
P4	Testing & Quality Assurance	11 Jun – 14 Jun	4 days
P5	Bug Fixing & Refinement	14 Jun – 15 Jun	2 days
P6	Documentation, Deploy & Submit	15 Jun – 16 Jun	2 days

2 Gantt Chart



3 Detailed Weekly Breakdown

Week 1 — Research & Requirements

18 – 24 May

- Define the problem: manual, spreadsheet-based depot route planning.

- Study existing fleet-management solutions and industry best practice.
- Identify target users and gather functional + non-functional requirements.
- Decide the technology stack (Laravel + Livewire + Flux UI).
- **Output:** Requirements Specification document.

Week 2 — Planning & System Design

25 – 31 May

- Project plan, task breakdown and this timeline.
- Database schema, ER diagram, use-case and architecture diagrams.
- Role / permission matrix and UI design system (app.css).
- Wireframe key screens; initialise repository and dev environment.
- **Output:** Design document + initialised codebase.

Weeks 3–4 — Core Development

01 – 11 Jun

- Authentication & authorisation (Fortify, 2FA, password confirmation).
- User management, roles, route & stop planning with interactive map.
- Vehicle / driver records; fuel & maintenance logs with service-due logic.
- Scheduling with conflict detection, trips board, dashboard & network map.
- Reporting engine with PDF export and activity / audit logging.
- **Output:** Feature-complete SRMSS application.

Testing & Quality Assurance

11 – 14 Jun

- Unit and feature tests with Pest; lint checks (Pint).
- Manual test checklist and role-based access-control verification.

Bug Fixing & Refinement

14 – 15 Jun

- Resolve defects found in testing; refine security settings & 2FA.
- Restrict profile / security settings access by role; final regression pass.

Documentation, Deployment & Submission

15 – 16 Jun

- Complete user guide, final report and diagrams.
- Prepare deployment package (InfinityFree) and deploy.
- Final review and coursework submission.

4 Key Milestones

M1

Requirements Specification approved

24 May 2026

M2

System Design & DB schema finalised

31 May 2026

M3

Auth + core modules functional

07 Jun 2026

M4

Feature-complete (all modules built)

M5

Testing & QA complete

M6

Project deployed & submitted

11 Jun 2026

14 Jun 2026

16 Jun 2026

5 Deliverables Summary

- Requirements Specification document **(M1)**
- System Design document + ER / use-case diagrams **(M2)**
- Working SRMSS web application **(M3, M4)**
- Automated test suite + manual test checklist **(M5)**
- User guide and final project report **(M6)**
- Deployed application + source-code repository **(M6)**

Note: This is a planned, industry-standard SDLC schedule presented for project documentation. Phase boundaries are indicative — some activities (e.g. testing and bug fixing) overlapped with development in practice.